

Refactor from
RT-300S → Zemax → FRED
to
RT-300S → FRED → Zemax

2026/03/21

Motivation

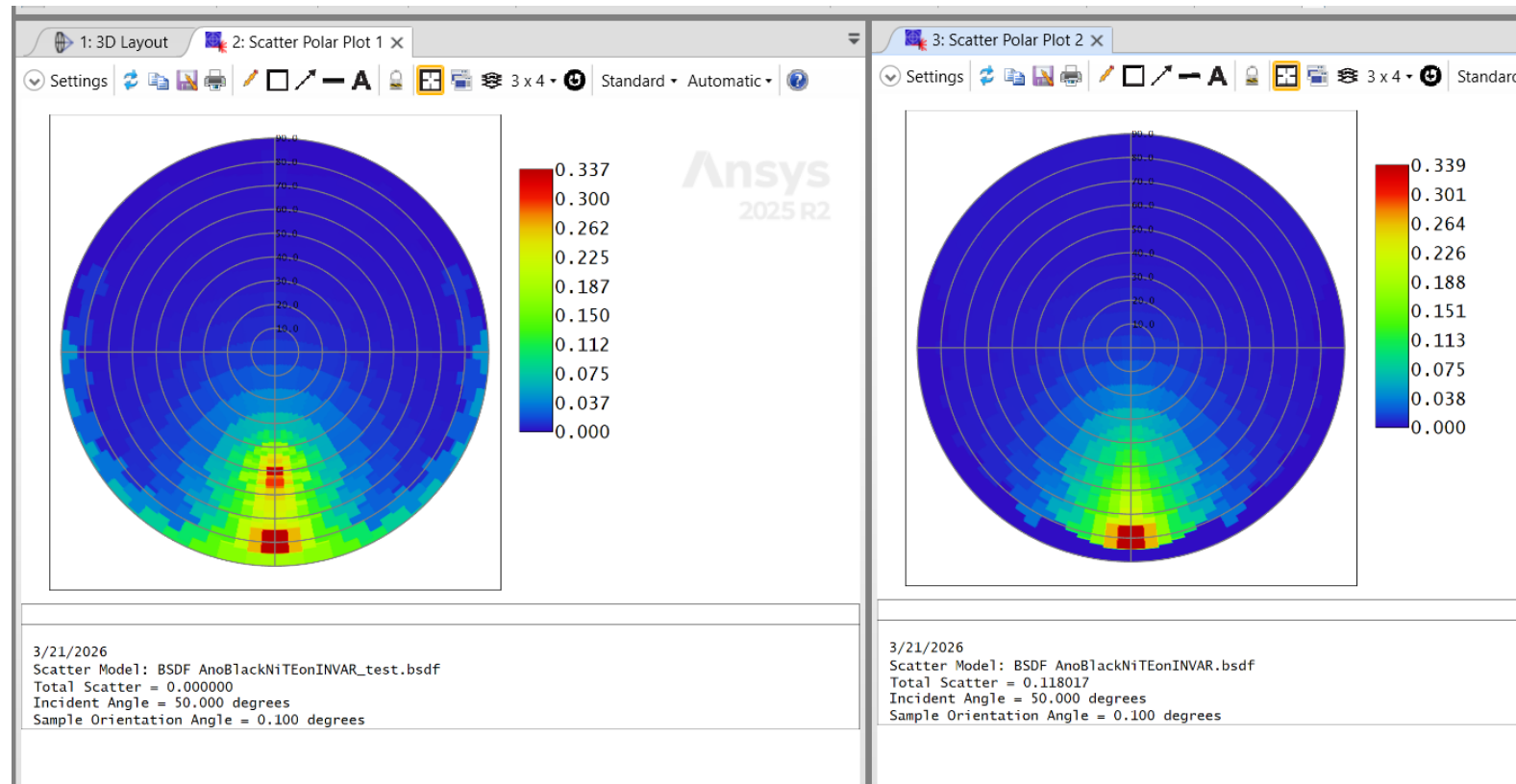
- In general, want code to be more modular (not single script(s), but various functions)
 - Makes far more readable and debuggable
- But, really want anisotropic in FRED, and other than cross-checking (in Zemax) the IMX455 data for RT-300S → Zemax → FRED with RT-300S → FRED → Zemax can RT-300S → FRED be confidently trusted, since RT-300S → Zemax is already trusted
 - Not sure how to validate Zemax → FRED for anisotropic, so this refactor bypasses that

Results

- Refactor seems to work well
 - All coordinate transformations seem fine
 - Zemax comparison is very similar
 - Main limitation currently is the zemax interpolation assumptions, but the goal here was to validate RT-300S → FRED, so improving that interpolation is a future work

Interpolation to Zemax grid

- Zemax interpolation assumptions still need to be worked out
 - Here, we see that assuming max BRDF as the specular value causes bug (left) at large incidence



Validation test of anisotropic FRED

- (1) To validate anisotropic RT-300S $\rightarrow$ FRED, doing the following:
 - (2) Anisotropic RT-300S $\rightarrow$ Zemax was considered “trusted enough” to move forward with
 - (3) Anisotropic FRED $\rightarrow$ Zemax is not yet validated
 - Then, if (1) $\rightarrow$ (3) yields similar result as (2), both (1) and (3) may be considered “trusted enough” to the degree that (2) is
 - And this increases trust that (2) is correct as well, because convergence on cross check

Validation test of anisotropic FRED

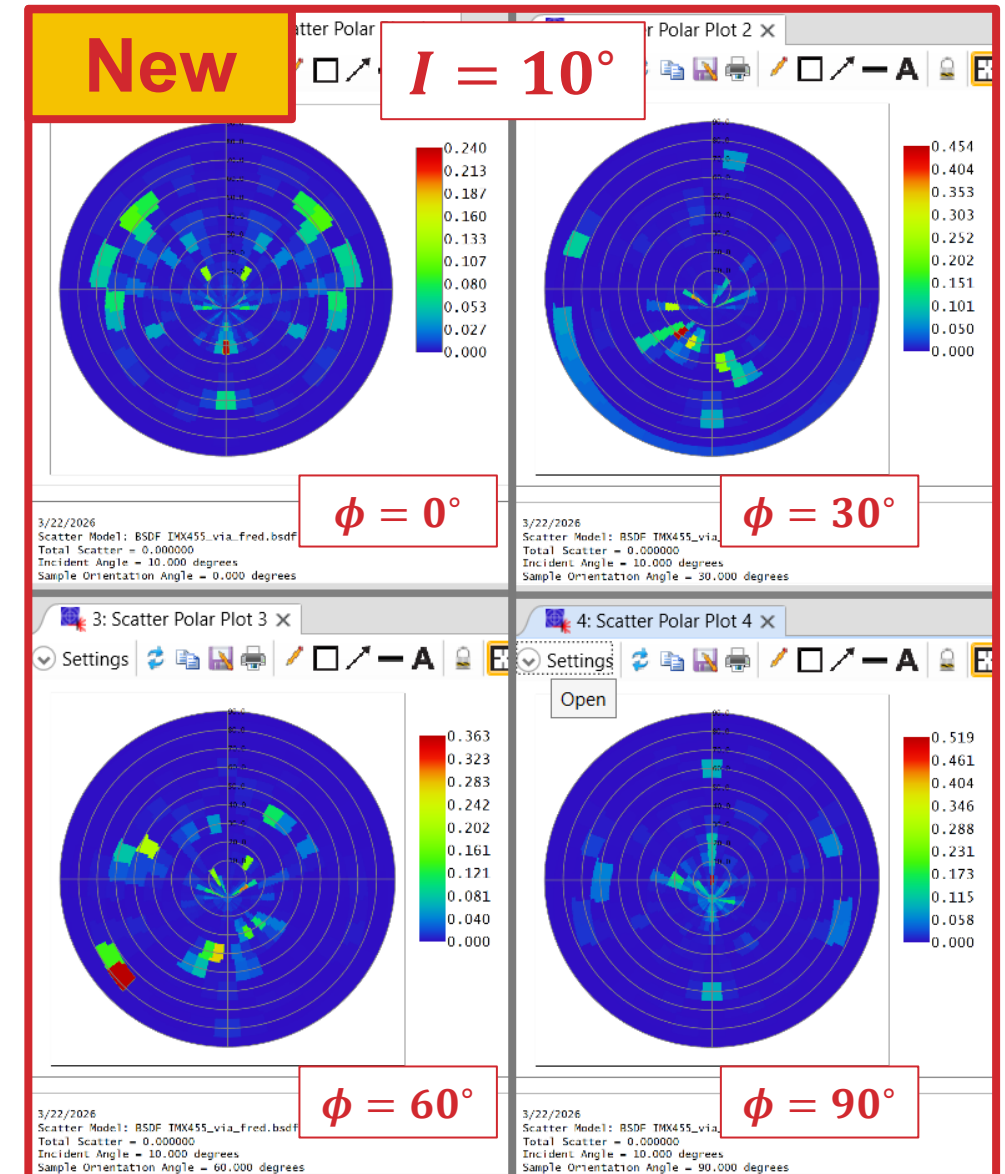
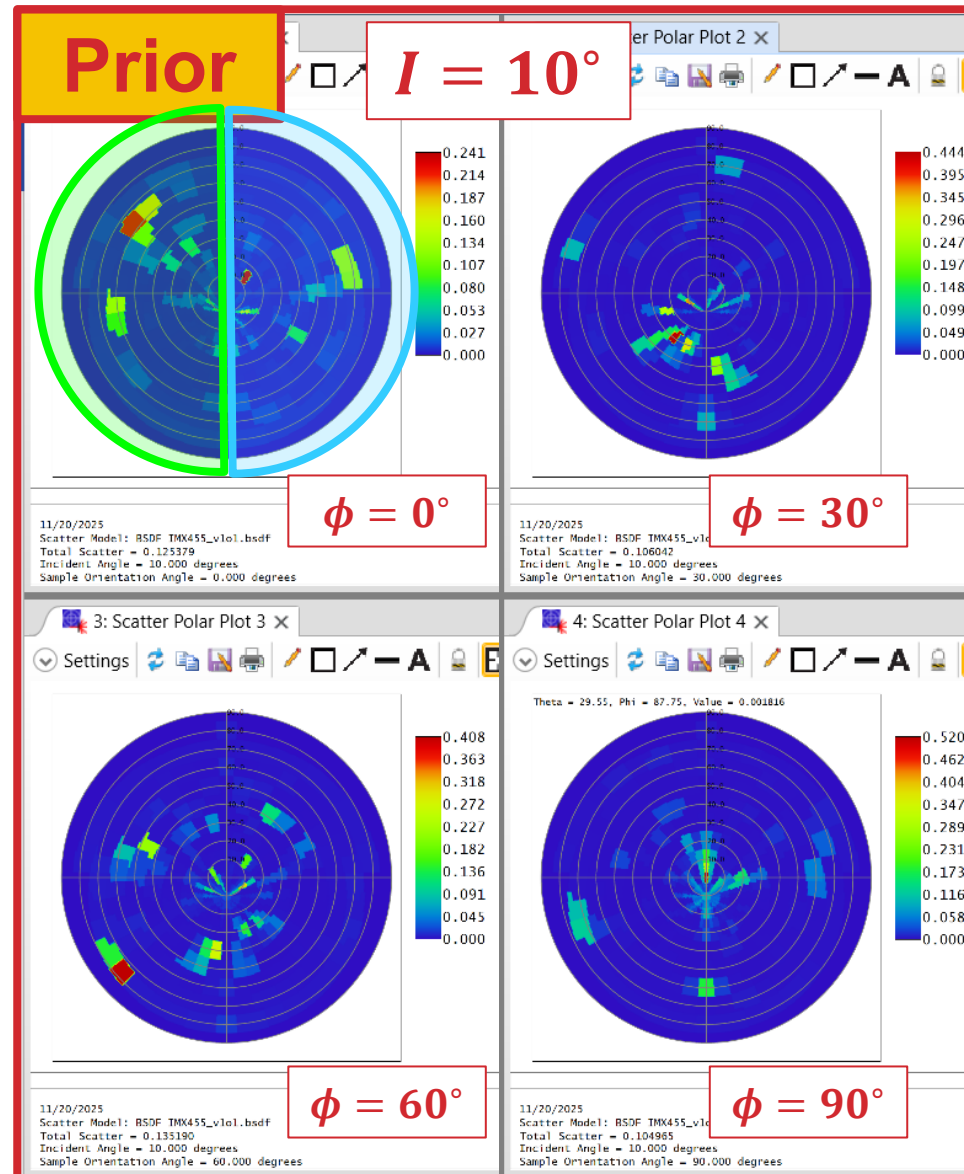
(recall prior IMX455 Zemax plots)

If anything, suggests bug in **Prior** at $\phi = 0^\circ$.

But, that is explainable because raw measurements capture full hemisphere, and RT-300S $\rightarrow$ Zemax may not have averaged across XZ.

But, because assuming the sample has symmetry about its XZ plane, averaging the (1) and (2) regions is intentional for **New**.

Perhaps $\phi = 0^\circ$ of **Prior** suggests the XZ symmetry assumption is poor for the IMX455, but that is a separate issue from the data processing validation test itself.



Validation test of anisotropic FRED

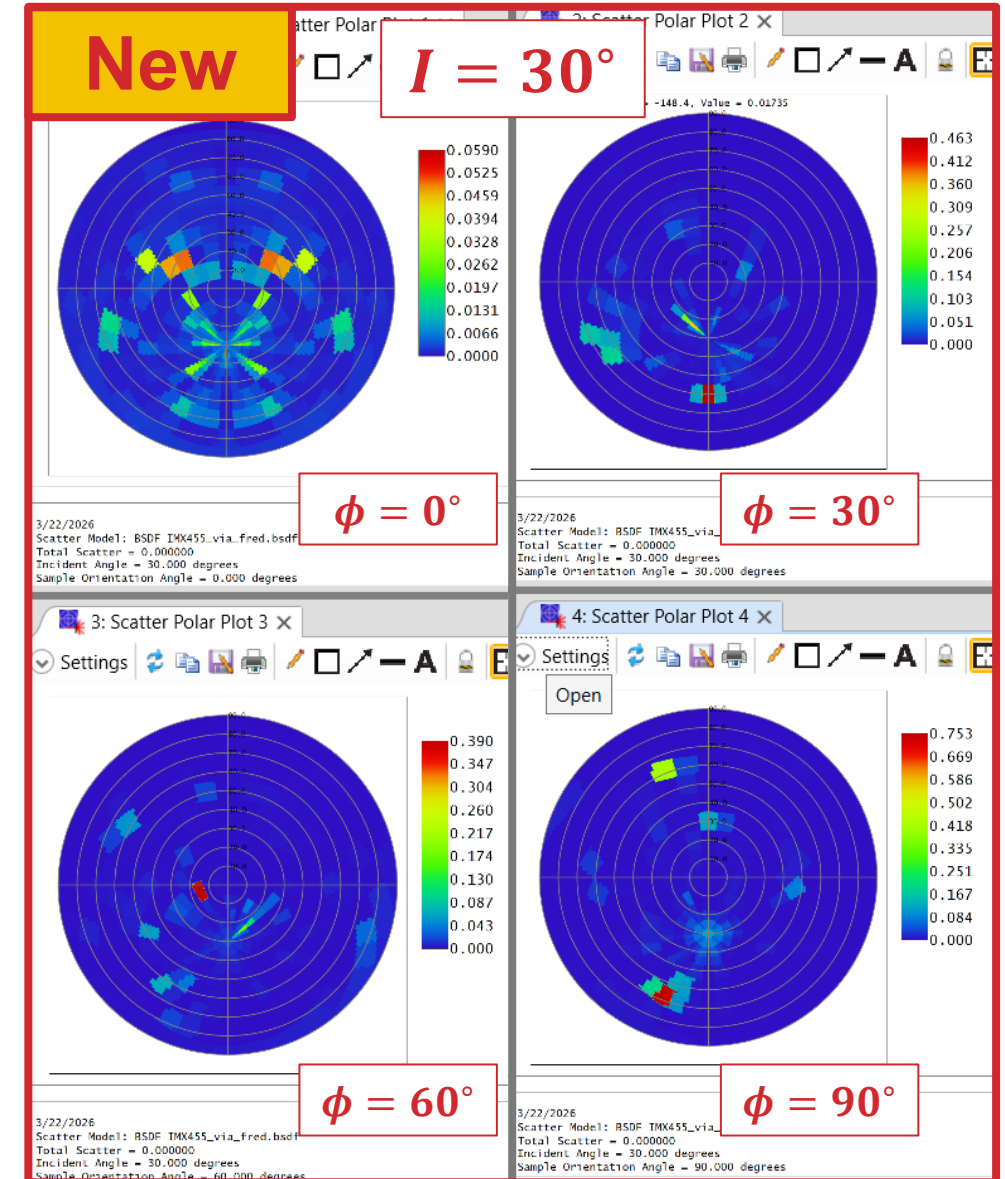
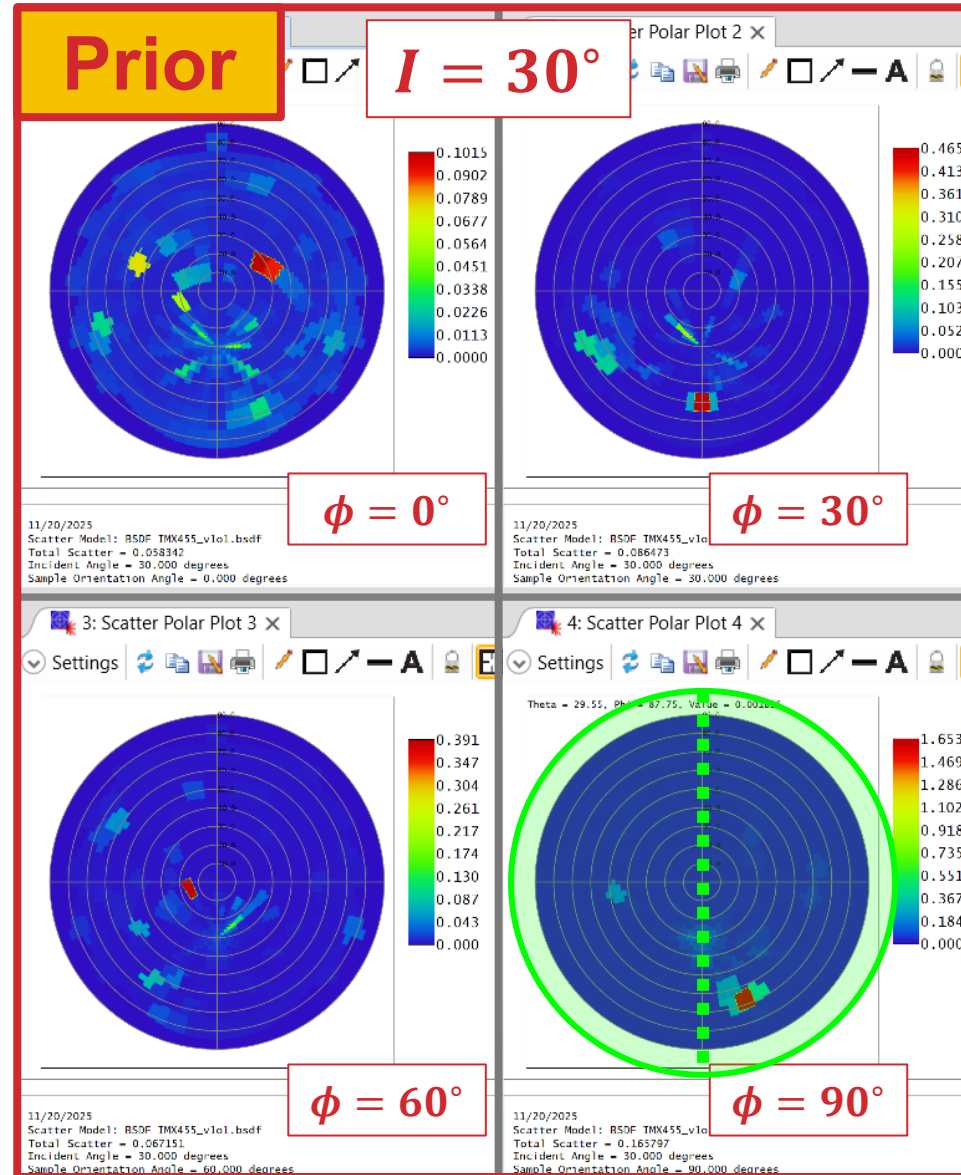
(recall prior IMX455 Zemax plots)

Same $\phi = 0^\circ$ concern.

But now, also concern about $\phi = 90^\circ$ **mirroring Prior $\rightarrow$ New.**

Meaning, $\phi_{\text{Prior}} = \phi_{\text{New}}$, except at 90° , where $\phi_{\text{Prior}} = -\phi_{\text{New}}$. Not sure why.

Since $\phi \in [0^\circ, 90^\circ)$ agrees, and “specular” peak movement as $\phi \rightarrow 90^\circ$ seems to agree with **New** pattern, concluding **New** is actually correction to **Prior**.

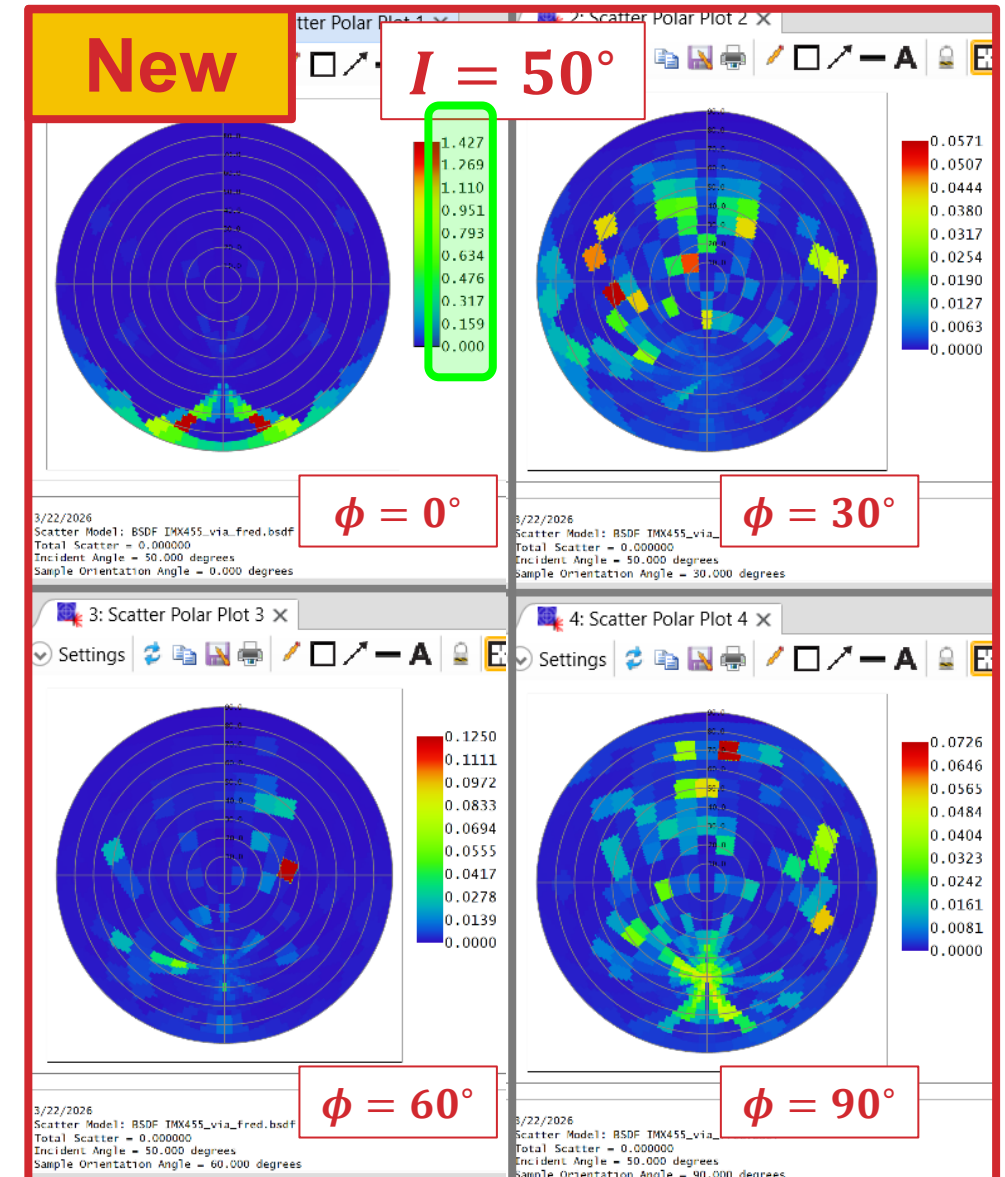
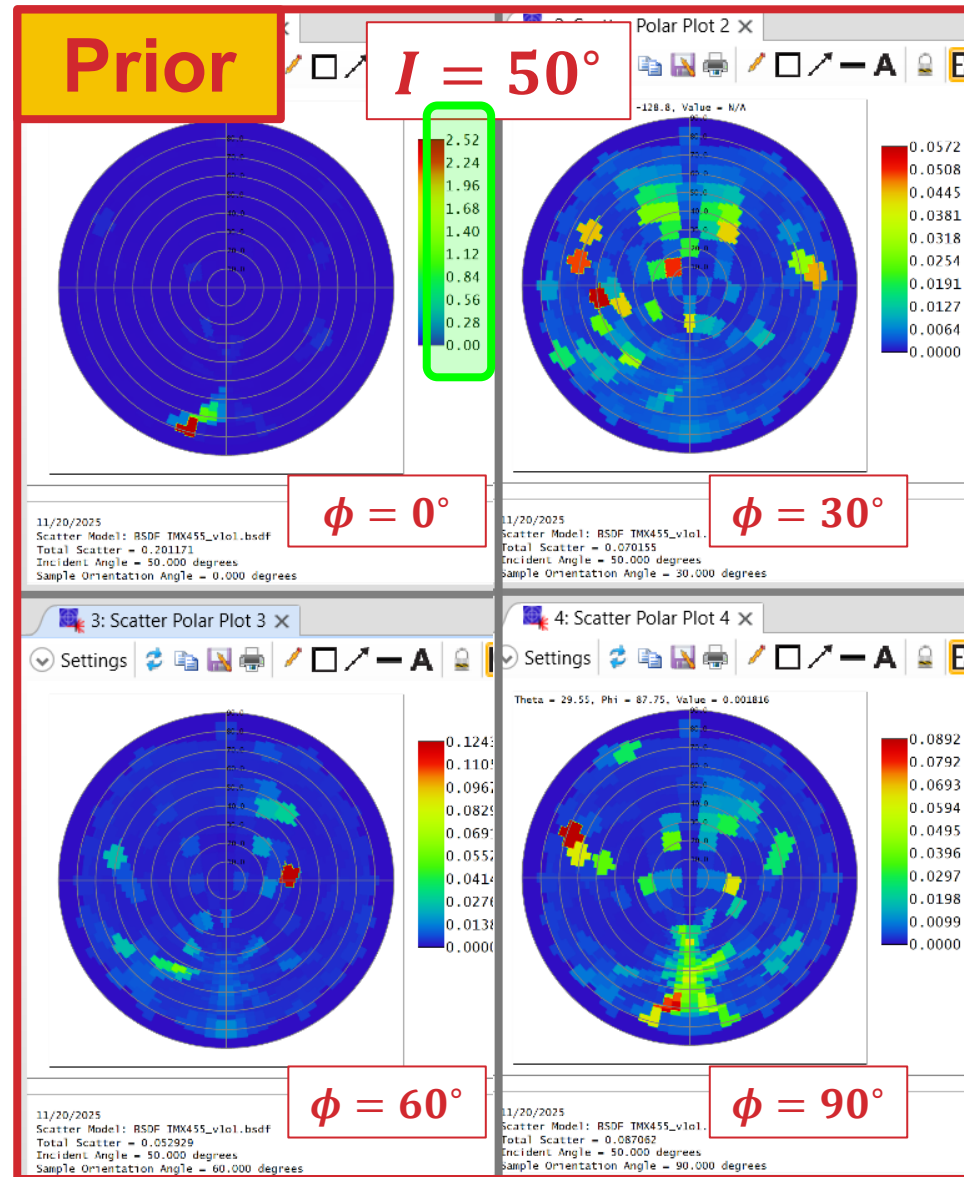


Validation test of anisotropic FRED

(recall prior IMX455 Zemax plots)

$\phi = 0^\circ$ here is good example of averaging across XZ assumption (again, separate issue from data processing itself). Note half BRDF scale.

Same $\phi = 90^\circ$ concern.

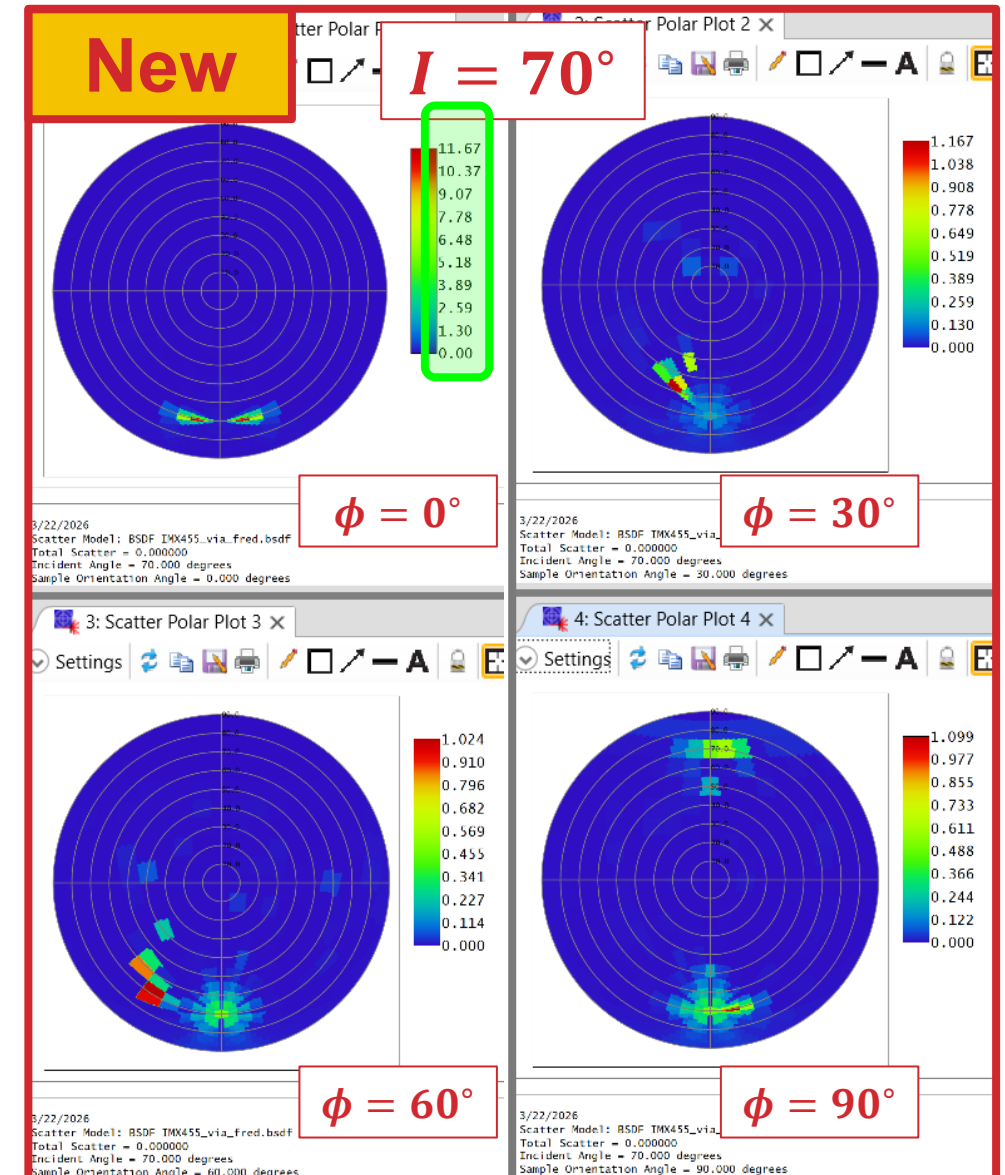
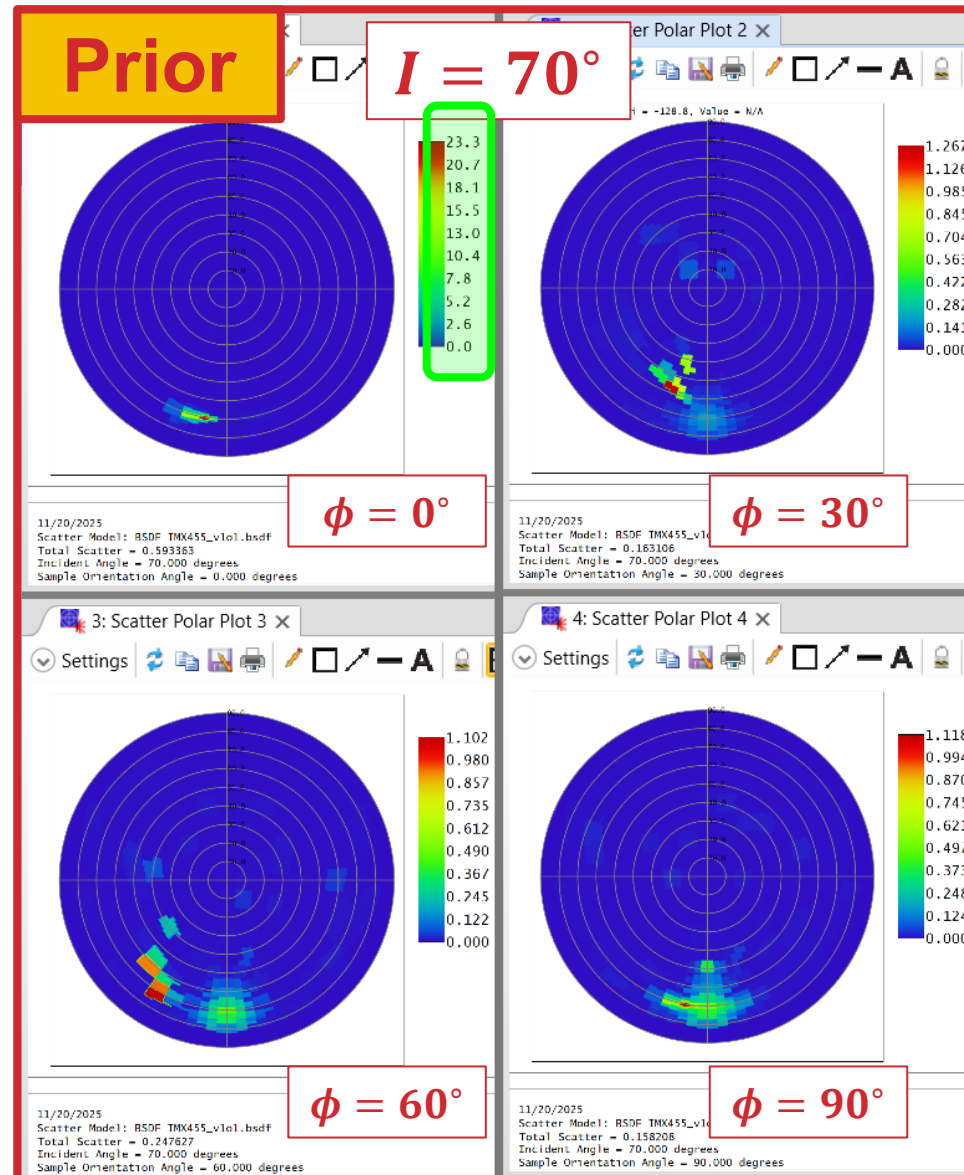


Validation test of anisotropic FRED

(recall prior IMX455 Zemax plots)

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Same $\phi = 90^\circ$ concern.



Validation test of anisotropic FRED

(recall prior IMX455 Zemax plots)

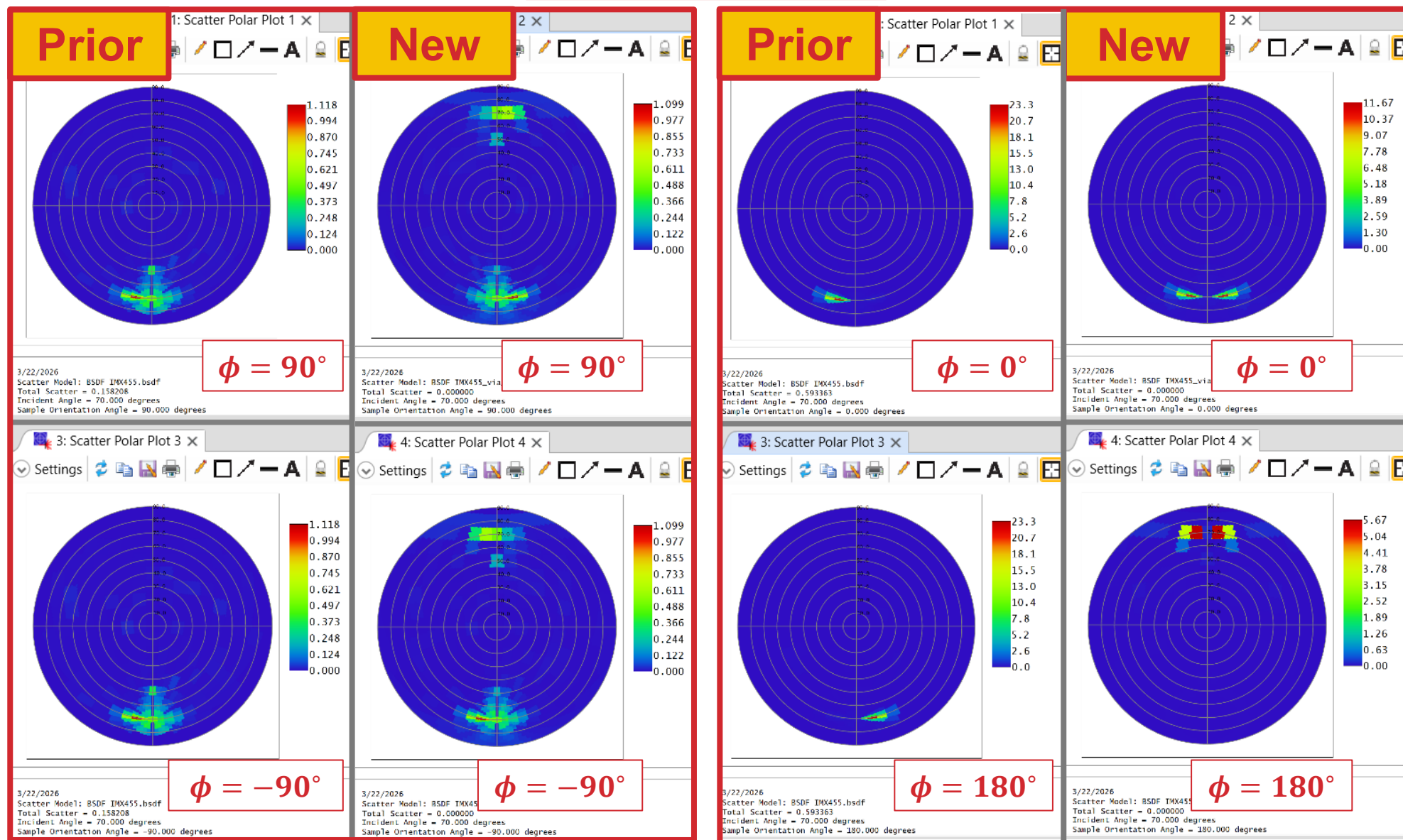
All at $I = 70^\circ$

Understanding the $\phi = 90^\circ$ concern:

Not obvious at first glance, but if visualizing $\phi = 0^\circ, 180^\circ$ of **New**, it indeed is the expected result of the XZ sample symmetry assumption. (imagine a cylindrical XZ-symmetric optic as the sample, with tilt about Y axis).

Similarly, $\phi = \pm 90^\circ$ of **New** is the expected results of YZ symmetry assumption.

Visualizing from perspective of source helps, with scatter cone plane (plotted) seen from global 3D view. May generate clearer visualization in future.



Validation test of anisotropic FRED

(recall prior IMX455 Zemax plots)

- **Conclusion:**

- Zemax results *qualitatively* align very well, so RT-300S → FRED is validated
- Explanation of discrepancies:
 - Quantitative differences may mostly be attributed to differences in the data processing's interpolation routine
 - Where results don't qualitatively align (namely, $\phi = 90^\circ n$ where $n \in 0, \pm 1, \pm 2, \dots$), we see a correction to unnoticed bugs in the RT-300S → Zemax routine, by visualizing what the expected results of XZ and YZ sample symmetry are, and seeing RT-300S → FRED → Zemax results match those expectations